

Euclid's Elements — Book XI

Proposition 8

Formal Reconstruction — Technical Documentation

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If two straight lines be parallel, and one of them be at right angles to any plane, the remaining one will also be at right angles to the same plane.

1 Introduction

Proposition XI.8 is the Euclidean converse companion to XI.6. Proposition XI.6 proved that two straight lines perpendicular to the same plane are parallel. Proposition XI.8 starts with two parallel spatial lines and assumes that one of them is perpendicular to a reference plane; the conclusion is that the second parallel is perpendicular to the same plane.

The proposition is also a structural boundary in the reconstruction. The formalizations XI.4–XI.7 did not require Hilbert Group IV. XI.8 does. The reason is already visible in Euclid's classical proof: the decisive transversal step uses Proposition I.29, the converse direction of the neutral parallel-angle criterion.

The Lean proof preserves the synthetic geometry but makes the Euclidean dependency more explicit. Instead of calling a planar I.29 wrapper, it separates the argument into

neutral I.27 + uniqueness of the parallel through a point.

The second component is precisely the plane-local Group IV assumption.

A second feature is genuinely three-dimensional. Euclid begins by saying that the two parallel lines meet the reference plane at points B and D . The first intersection is given by the hypothesis that the first line is perpendicular to the plane. The second intersection is not automatic in an abstract incidence space. The formal proof derives it.

2 Euclid's Statement

Euclid's Proposition XI.8 states:

If two straight lines be parallel, and one of them be at right angles to any plane, the remaining one will also be at right angles to the same plane.

Let AB and CD be parallel straight lines, and suppose that AB is perpendicular to the reference plane π at B . The claim is that CD meets π at a point D and is perpendicular to π there.

3 Classical Configuration

Euclid lets the parallel lines AB and CD meet the reference plane at B and D . He joins BD , draws DE in the reference plane perpendicular to BD , makes $DE = AB$, and joins BE , AE , and AD .

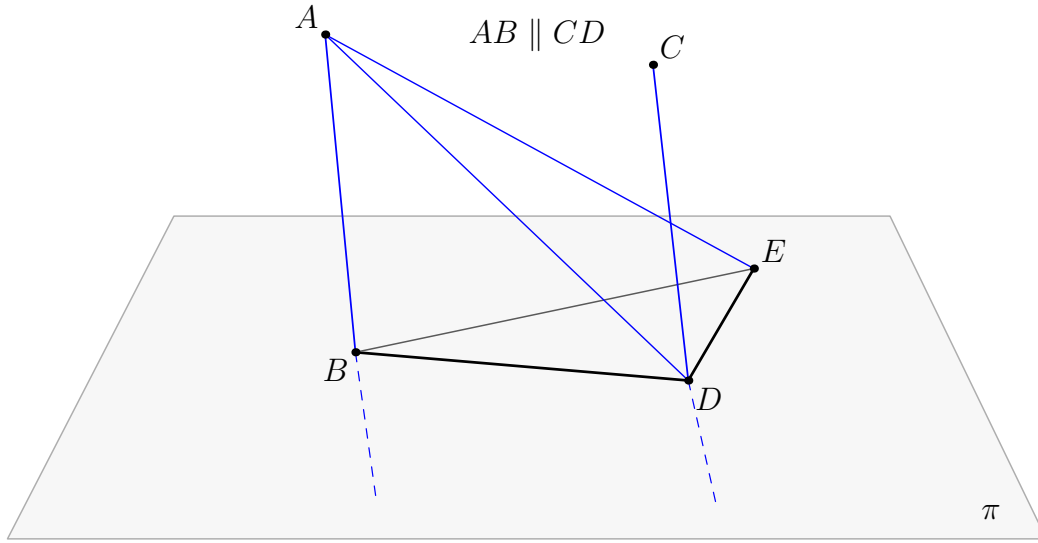


Figure 1: The classical configuration of Proposition XI.8. Black lines lie in the reference plane π . Blue lines or segments leave π . The hidden continuations of the two spatial parallel carriers are shown blue and dashed. The auxiliary construction reproduces the metric core already used in XI.6: $DE \perp BD$ and $DE = AB$.

The line BD plays two roles. It is a line of the reference plane and also lies in the common plane of the spatial parallels. This is the transversal on which the classical I.29 step acts.

4 Classical Proof

The local classical proof has five main stages.

First, since $AB \parallel CD$, Proposition XI.7 places AB , CD , and the joining line BD in one plane. The opening phrase that the parallel lines meet the reference plane at B, D is diagrammatic in the classical text.

Second, Euclid constructs DE in the reference plane with

$$DE \perp BD, \quad DE = AB.$$

Since $AB \perp \pi$, the angles

$$\angle ABD, \quad \angle ABE$$

are right.

Third, BD falls on the parallel lines AB and CD . Proposition I.29 says that the appropriate interior angles sum to two right angles. Since $\angle ABD$ is right, $\angle CDB$ is right. Hence

$$CD \perp BD.$$

Fourth, Euclid repeats the metric triangle argument familiar from XI.6. From

$$AB = DE, \quad BD = DB,$$

and equality of the included right angles, Proposition I.4 gives

$$AD = BE.$$

Then

$$AB = DE, \quad BE = AD, \quad AE = EA,$$

so Proposition I.8 gives

$$\angle ABE = \angle EDA.$$

The former is right, hence the latter is right:

$$ED \perp AD.$$

Now ED is perpendicular to the two intersecting lines DB and DA . Proposition XI.4 therefore gives

$$ED \perp \text{plane}(BDA).$$

The line CD lies in this plane, so

$$ED \perp CD.$$

Finally, in the reference plane π the two lines DB and DE meet at D , and CD is perpendicular to both:

$$CD \perp DB, \quad CD \perp DE.$$

A second application of XI.4 yields

$$CD \perp \pi.$$

The classical local dependency chain is therefore

$$\begin{array}{c}
AB \parallel CD, \quad AB \perp \pi \\
\Downarrow \\
\text{XI.7} \quad \text{and the joining line } BD \\
\Downarrow \\
DE \perp BD, \quad DE = AB \quad (\text{I.11, I.3}) \\
\Downarrow \\
AB \perp \pi \Rightarrow \angle ABD, \angle ABE \text{ right} \\
\Downarrow \\
\text{I.29} \\
\Downarrow \\
CD \perp BD \\
\Downarrow \\
\text{I.4, I.8} \\
\Downarrow \\
ED \perp AD \\
\Downarrow \\
\text{XI.4} \\
\Downarrow \\
ED \perp CD \\
\Downarrow \\
\text{XI.4} \\
\Downarrow \\
CD \perp \pi.
\end{array}$$

The incidence assertion that CD belongs to the plane through BD, DA also uses the fact that the triangle ABD is planar and that AB, CD, BD share the plane supplied by XI.7. In a formal proof this plane identification cannot be left to the printed diagram.

5 What is genuinely three-dimensional here?

There are two distinguished planes.

The first is the reference plane π , to which AB is perpendicular. The second is the common plane σ containing the parallel spatial lines AB and CD . Since AB is not contained in π , the planes are distinct.

The points B and D lie in both planes. Consequently their joining line

$$d = BD$$

lies in both π and σ ; in the formal proof it is the relevant line of intersection of the two plane slices.

The auxiliary line

$$e = DE$$

lies in π , while the line

$$a = DA$$

lies in σ . The proof therefore alternates between two induced plane geometries.

This change of plane is not bookkeeping. It is the spatial geometry of the proposition.

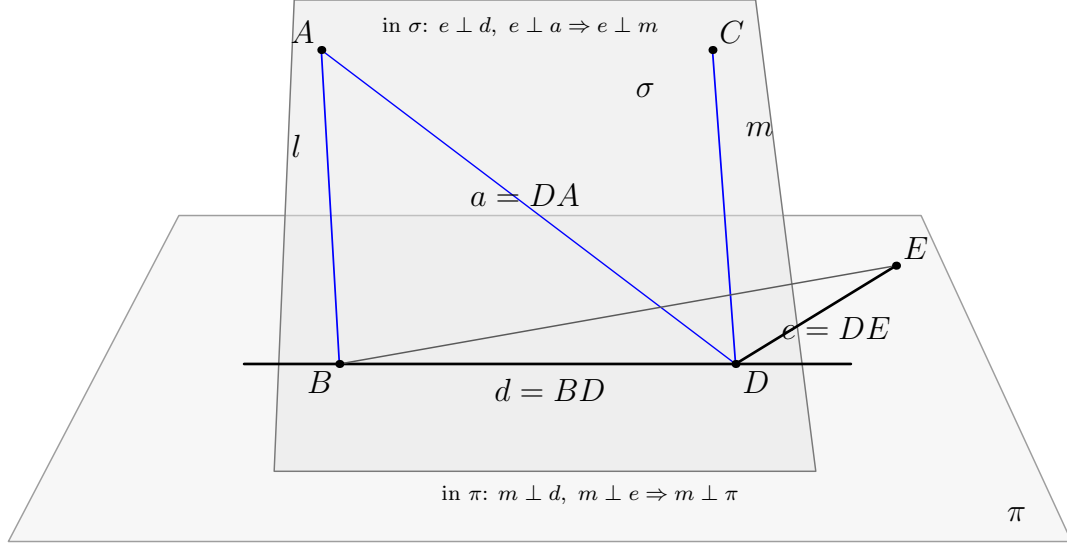


Figure 2: The two plane slices used by the reconstruction. The common line $d = BD$ lies in both π and σ . The first XI.4 application uses the two lines $d, a \subset \sigma$ to prove $e \perp \sigma$, hence $e \perp m$. The final XI.4 application uses $d, e \subset \pi$ to prove $m \perp \pi$.

6 Formal Statement

The production theorem is:

```

theorem euclid_proposition_11_8
  [H : HilbertIncidence Geo]
  [HilbertPlaneIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  [HSI : HilbertSpaceIncidence Geo]
  [_HSO : HilbertSpaceOrder
    (Geo := Geo) (H := H) (S := S)]
  [HSC : HilbertSpaceCongruence
    (Geo := Geo) (H := H) (S := S)]
  [HSE : HilbertSpaceEuclidean Geo]
  (l m : Geo.Line)
  (pi : S.Plane)
  (B : Geo.Point)
  (hParallel : HilbertSpaceLinesParallel Geo l m)
  (hPerp : HilbertLinePerpendicularPlaneAt Geo l pi B) :
  exists D : Geo.Point,
    HilbertLinePerpendicularPlaneAt Geo m pi D

```

The point D is deliberately not an input. Its existence is part of the theorem. This is stronger as a formal interface than assuming in advance that the second parallel already meets the plane.

7 Spatial / Planar Architecture Used

The ambient spatial layer consists of

```
HilbertSpacePrimitive
```

```

HilbertSpaceIncidence
HilbertSpaceOrder
HilbertSpaceCongruence
HilbertSpaceEuclidean
HilbertSpaceLinesParallel
HilbertLinePerpendicularPlaneAt
HilbertLinesPerpendicularAt

```

The proof then uses induced planar geometry in two different fixed planes:

$$\text{PlaneGeo}(\text{Geo}, \sigma) \quad \text{and} \quad \text{PlaneGeo}(\text{Geo}, \pi).$$

Inside σ , the proof uses neutral planar order, right angles, and the I.27 alternate-angle criterion. Inside π , the final configuration supplies two distinct intersecting lines d, e for XI.4.

The plane-space bridges used in the argument include

```

planeGeo_linesPerpendicularAt_iff_ambient
planeGeo_rightAngle_iff_ambient
planeGeo_angleCongruent_iff_ambient
planeGeo_not_primCollinear_to_ambient
planeGeo_linesDisjoint_iff_ambient

```

The last bridge was added with the Group IV layer. It identifies disjointness of two lines in a fixed `PlaneGeo` with ambient disjointness of their underlying spatial carriers.

8 The Euclidean Layer: Group IV

XI.8 is the first proposition in the present Book XI sequence whose formal proof requires the Euclidean parallel axiom.

The spatial axiom is not installed as a global `HilbertEuclideanPlaneGeo` instance. Such an instance would incorrectly treat the ambient three-space as a plane. Instead the axiom is stated plane-locally:

```

class HilbertSpaceEuclidean
  (Geo : Geometry.Geo)
  [H : HilbertIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  [HilbertSpaceIncidence Geo]
  [HilbertSpaceOrder Geo]
  [HilbertSpaceCongruence Geo] : Prop where

parallel_unique_in_plane :
  forall pi : S.Plane,
    forall l : Geo.Line,
      HilbertLineInPlane Geo l pi ->
        forall A : Geo.Point,
          S.OnPlane A pi ->
            Not (H.OnLine A l) ->
              forall b c : Geo.Line,
                HilbertLineInPlane Geo b pi ->

```

```

HilbertLineInPlane Geo c pi ->
H.OnLine A b ->
HilbertLinesDisjoint Geo b l ->
H.OnLine A c ->
HilbertLinesDisjoint Geo c l ->
b = c

```

For every fixed ambient plane, the bridge instance

```

instance planeGeoHilbertEuclidean
...
(pi : S.Plane) :
HilbertEuclideanPlane (PlaneGeo Geo pi)

```

recovers the existing planar Group IV API. The architectural rule is therefore preserved:

Euclidean parallelism is planar, even inside three-space.

9 Proof Architecture of the Reconstruction

9.1 Step 1: derive the hidden intersection point D

The helper

```

hilbert_XI8_parallel_meets_perpendicular_plane

```

proves that the second parallel m meets the reference plane.

Unpacking

$\text{HilbertSpaceLinesParallel}(l, m)$

gives a common plane σ containing l, m , with the two lines disjoint. The foot B of $l \perp \pi$ lies in both π and σ .

The planes are distinct. If $\sigma = \pi$, then $l \subset \pi$, contradicting the theorem

```

hilbert_linePerpendicularPlaneAt_not_in_plane

```

The general plane-intersection theorem then supplies a line

$$q = \pi \cap \sigma$$

through B .

Suppose m did not meet q . Then both l and q would be lines through B in σ disjoint from m . Group IV uniqueness would force $l = q$, but $q \subset \pi$, again contradicting $l \not\subset \pi$.

Hence m meets q at some point D , and therefore

$$D \in m \cap \pi.$$

This is the formal proof of a fact Euclid leaves implicit at the opening of XI.8.

9.2 Step 2: reuse the XI.6 metric core

With D fixed, the proof reuses

```
hilbert_XI6_second_SSS_right_angle
```

from Proposition XI.6.

The helper constructs points and lines corresponding to Euclid's auxiliary configuration and returns, in particular,

$$d = BD, \quad e = DE,$$

with

$$d, e \subset \pi, \quad e \perp d,$$

and the right angle

$$\angle EDA.$$

Thus the SAS/SSS part of XI.8 is not duplicated. The earlier XI.6 development is used as reusable spatial congruence infrastructure.

9.3 Step 3: first application of XI.4 in σ

Let a be the line DA . The proof shows

$$d, a \subset \sigma, \quad d \neq a.$$

The inequality is geometrically meaningful: A lies on the spatial line l but outside the reference plane π , while $d \subset \pi$.

The XI.6 metric result gives

$$e \perp d, \quad e \perp a.$$

Therefore XI.4, applied with the induced geometry of σ , gives

$$e \perp \sigma.$$

Since $m \subset \sigma$ and $D \in m$,

$$e \perp m.$$

This is the formal counterpart of Euclid's first XI.4 step.

9.4 Step 4: replace I.29 by neutral I.27 plus Group IV

The remaining task is to prove

$$m \perp d.$$

The proof does not call a public I.29 theorem. Instead, inside $\text{PlaneGeo}(\text{Geo}, \sigma)$, it constructs a line p through D perpendicular to d .

The neutral helper

```
hilbert_XI8_coplanar_perpendiculars_to_same_line_disjoint
```


shows that p and l are disjoint. Its proof is an I.27 argument: two right angles are normalized to equal alternate angles and the neutral alternate-angle theorem produces nonintersection.

But m is also disjoint from l , because $l \parallel m$. Thus m and p are two lines of the same plane σ , both passing through D , both disjoint from l . Group IV uniqueness forces

$$m = p.$$

Since $p \perp d$,

$$m \perp d.$$

This factorization isolates the exact Euclidean content:

$$\underbrace{\text{right angles} \Rightarrow \text{nonintersection}}_{\text{neutral I.27}} + \underbrace{\text{unique parallel through } D}_{\text{Group IV}} \implies m \perp d.$$

It is the formal counterpart of Euclid's I.29 step.

9.5 Step 5: final application of XI.4 in π

At this point

$$d, e \subset \pi, \quad m \perp d, \quad m \perp e.$$

The lines d and e are distinct because $e \perp d$.

The final call to

```
euclid_proposition_11_4
```

uses d, e as the two distinct intersecting lines of the reference plane. It returns

$$m \perp \pi$$

at D , which is the required conclusion.

10 Lean Formalization

The proposition-local proof is organized around four named helpers:

```
hilbert_XI8_parallel_meets_perpendicular_plane
hilbert_XI8_auxiliary_line_perpendicular_to_parallel
hilbert_XI8_coplanar_perpendiculars_to_same_line_disjoint
hilbert_XI8_parallel_line_perpendicular_to_given_BD
```

The actual Lean DAG is

$$\begin{array}{c}
l \parallel m, \quad l \perp \pi \text{ at } B \\
\Downarrow \\
\text{hilbert_XI8_parallel_meets_perpendicular_plane} \\
\Downarrow \\
D \in m \cap \pi \\
\Downarrow \\
\text{hilbert_XI6_second_SSS_right_angle} \\
\Downarrow \\
d = BD, \quad e = DE, \quad e \perp d, \quad \angle EDA \text{ right} \\
\Downarrow \\
\text{XI.4 in } \sigma \\
\Downarrow \\
e \perp m \\
\Downarrow \\
\text{I.27 in PlaneGeo}(\sigma) + \text{Group IV} \\
\Downarrow \\
m \perp d \\
\Downarrow \\
\text{XI.4 in } \pi \\
\Downarrow \\
\text{euclid_proposition_11_8.}
\end{array}$$

The production file imports XI.4, XI.5, and XI.6. It does not import XI.7. The common plane that Euclid obtains through XI.7 is already contained in the formal hypothesis `HilbertSpaceLinesParallel`.

11 Hidden Formal Data

XI.8 contains substantially more hidden incidence information than its classical diagram suggests.

- The second parallel must be proved to meet the reference plane. The point D is an existential output, not an assumed input.
- The common plane σ of the parallels must be proved distinct from π .
- The intersection line $q = \pi \cap \sigma$ is constructed explicitly.
- The points B, D must be proved distinct from disjointness of the parallel carriers.
- The line $d = BD$ must be shown to lie both in π and in σ .
- The point A selected on l must be proved to lie outside π ; otherwise the line l would lie in π , contrary to $l \perp \pi$.
- The line $a = DA$ must be constructed and shown distinct from d .
- Right-angle and perpendicularity statements must be transported between ambient geometry and `PlaneGeo`.
- The neutral I.27 argument requires explicit `SameSide` / `OppositeSide` and `SameRay` normalization that Euclid suppresses in the diagram.

- The two disjointness orientations required by Group IV are supplied explicitly.
- Before the final XI.4 application, $d \neq e$ is derived from $e \perp d$.

Subtype projections and record fields occur in the Lean code, but they are implementation details. The mathematically significant hidden data are the plane memberships, distinctness statements, side orientations, and the derived intersection point D .

12 Relationship with Earlier Book XI Propositions

The classical and formal dependency graphs differ in a controlled way.

Classically, XI.7 is an explicit opening dependency: it places AB, CD, BD in one plane. XI.4 is then used twice. The classical plane-identification step also relies on the ordinary fact from XI.2 that a triangle lies in one plane.

The production Lean file imports

```
import CGJteamLab.Proposition11_4
import CGJteamLab.Proposition11_5
import CGJteamLab.Proposition11_6
```

but does not import Proposition XI.7.

XI.4 is used twice exactly as in the classical architecture: first in the common plane σ , then in the reference plane π .

XI.5 contributes established line–plane perpendicularity infrastructure, including the theorem that a line perpendicular to a plane is not contained in that plane.

XI.6 contributes the reusable metric construction and the spatial SAS/SSS right-angle machinery. This reuse is substantial: the central auxiliary configuration of XI.8 is almost the same as the configuration built for XI.6.

XI.7 is absent from the Lean import graph because `HilbertSpaceLinesParallel` already stores the common carrier plane as part of the definition of spatial parallelism. Thus the formal proof consumes the conclusion-like data that XI.7 exposes, without invoking the proposition theorem itself.

13 Relationship with Book I / Book Zero / Hilbert Infrastructure

The classical proof uses I.11 and I.3 to construct the auxiliary perpendicular and equal segment. In the Lean reconstruction these operations are already packaged in the XI.6 helper chain over the Hilbert/Book Zero construction interface.

The metric part corresponds to I.4 and I.8 and is likewise inherited from XI.6.

The parallel step is the important logical boundary. The direction represented by I.27,

$$\text{equal alternate angles} \implies \text{nonintersection},$$

belongs to the neutral congruence/order layer. The reverse use of parallelism represented by I.29 requires the Euclidean parallel axiom. The formal proof displays this by using the neutral I.27 theorem first and applying Group IV only to identify the unique line through D disjoint from l .

No coordinate system, vector space, scalar product, distance formula, trigonometric function, or numerical angle measure is used.

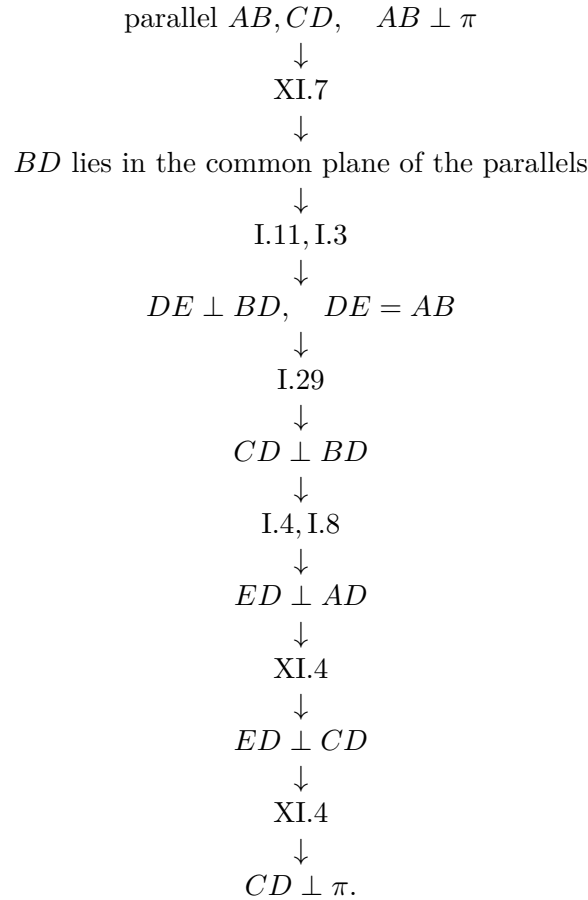
14 Relationship with the Foundations

XI.8 is the first proposition in this local sequence whose current Euclid-faithful reconstruction genuinely uses Hilbert Group IV. The public theorem assumes **HilbertSpaceEuclidean Geo**, and the proof consumes that structure through the induced Euclidean geometry of an explicit plane slice.

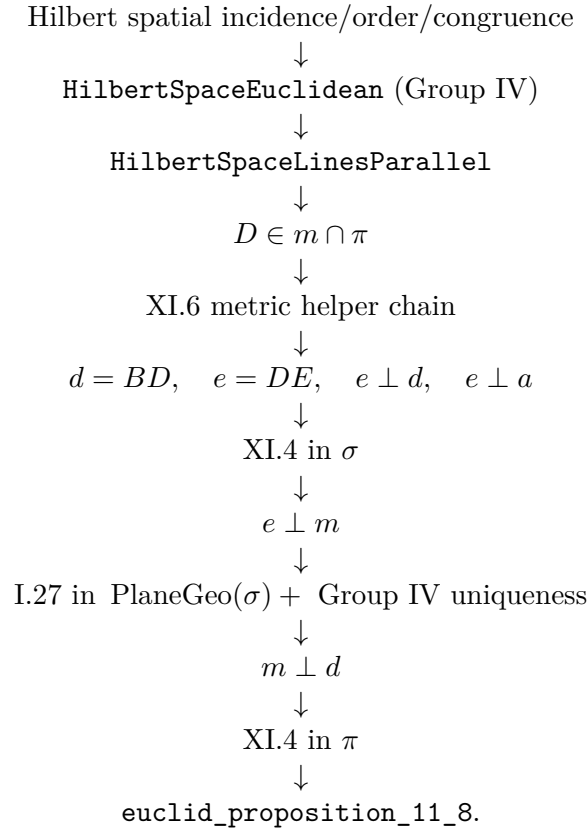
This use of Group IV is already analyzed in the section “The Euclidean Layer: Group IV” above. The essential point is mathematical: the proof needs the uniqueness/existence behavior of parallels in the relevant plane; no global ambient planar instance is installed. No continuity principle is used.

15 Dependency Summary

The classical local chain is



The actual formal chain is



The dependency classification is therefore:

New proposition-specific global axiom	no
New spatial axiom class	yes: HilbertSpaceEuclidean
Continuity	no
Group IV / Euclidean parallelism	yes
Earlier Book XI production imports	XI.4, XI.5, XI.6
Classical explicit Book XI dependency	XI.7, XI.4
New PlaneGeo Euclidean bridge	yes
Coordinates / vectors / scalar product	no
sorry/admit	no

16 Conclusion

Proposition XI.8 is the first point in the present Book XI reconstruction where Euclidean parallel uniqueness is mathematically necessary. The formal proof makes that boundary sharper than the classical text: neutral I.27 proves nonintersection, while Group IV supplies the uniqueness needed to recover the I.29 direction.

At the same time, the proof remains fully synthetic. It reuses the metric construction of XI.6, applies XI.4 in two different plane slices, and derives the hidden intersection point of the second parallel with the reference plane instead of assuming it from the diagram.

The result therefore tests three parts of the spatial architecture at once: plane-local Euclidean

parallelism, transport between ambient three-space and induced plane geometry, and the universal line–plane perpendicularity predicate. No continuity or analytic representation is introduced.